COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	SECTION: CPE21S4	
DATE PERFORMED:	31/07/2026	DATE SUBMITTED:	31/07/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

ILOB2 > main.cpp > ...

```cpp
#include<iostream>
#include "stackList.h"

int main(){

    std::cout<< " --- PUSHING 'c' 'p' 'e' ONTO THE STACK ---- \n";
    //push
    push('c');
    push('p');
    push('e');

    //display the current top
    Top<char>(); // Top should be 'e'
    displayALL<char>();
    std::cout<<"\n ------- Popping all the elements ------- \n";

    while(!isEmpty<char>()){
        std::cout<< " Popped: "<< pop<char>() <<std::endl;
    }

std::cout<<"\n ------- Testing the Underflow ------- \n";
pop<char>();

return 0;
}
```

ILOB2 > C stacklist.h > ...

```
#ifndef STACK_LIST_H
#define STACK_LIST_H

//class node that contains data and the next
template <typename T>
class Node{
public:
T data; // VALUE THAT WE NEED STORE
Node* next; // POINTER TO THE NEXT NODE
};

//global pointer for the top of the stack
template <typename T>
Node<T> *head = nullptr;

//operations for the stack

// push
template <typename T>
void push(T newData){
//create a new node and assign the newData
Node<T>* newNode = new Node<T>;
newNode->data = newData;

//new node points to current top, then becomes the new top
newNode->next = head<T>;
head<T> = newNode;
}

// pop
template <typename T>
T pop(){
//check if the stack is empty:
if(head<T> == nullptr){
std::cout<< " stack overflow. "<<std::endl;
}

//save the current top node
Node<T>* temp = head<T>;
```

```
40    //store its data
41    T tempVal = temp->data;
42    // move the head to the next node
43    head<T> = head<T>->next;
44    //free the removed node
45    delete temp;
46    //return the removed value
47    return tempVal;
48    }
49
50    // isEmpty
51    template<typename T>
52    bool isEmpty(){
53    return head<T> == nullptr;
54    }
55
56    // top
57    template<typename T>
58    void Top(){
59    // error catching:
60    if(isEmpty<T>()){
61    std::cout<<"Stack underflow "<<std::endl;
62    return;
63    }
64    // display the Top node
65    std::cout<<"Top of Stack: "<<head<T>->data <<std::endl;
66    }
67
68    //display All
69    template <typename T>
70    void displayALL(){
71    if (head<T> == nullptr){
72    std::cout << " stack undeflow\n";
73    return;
74    }
75    std::cout << "Displaying the stack: \n";
```

```
75    std::cout << "Displaying the stack: \n";
76    Node<T>* temp = head<T>;
77    while(temp != nullptr){
78    std::cout << temp->data << " \n";
79    temp = temp->next;
80    }
81    std::cout << "\n";
82    }
83
84    #endif
```

```
PS C:\Users\TIPQC\Documents\CPES4_Sevilla\week 4\week4.1>  & '
auncher.exe' '--stdin=Microsoft-MIEngine-In-krw1da1a.1tk' '--s
crosoft-MIEngine-Pid-a53ffjr3.tav' '--dbgExe=C:\msys64\ucrt64\
 --- PUSHING 'c' 'p' 'e' ONTO THE STACK ----
Top of Stack: e
Displaying the stack:
e
p
c


 ------- Popping all the elements -------
 Popped: e
 Popped: p
 Popped: c

 ------- Testing the Underflow -------
 stack overflow.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Copilot

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: I used it for helping me to fix my code to work on (Programiz)

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: It help me to fit the code on Programiz website because my VScode still not working.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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Supplementary: